

GHG Emissions Inventory Artisanal Fishing Fleet Mexican Pacific Coast | 2022

GHG Protocol Corporate Standard | IPCC AR6 GWP100 | Scopes 1, 2 & 3 Tier 2
Approximation based on publicly available data

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Tools: R (tidyverse · ggplot2)

Executive Summary

This report presents a Greenhouse Gas (GHG) emissions inventory for the artisanal fishing fleet operating along the Mexican Pacific coast, covering Sinaloa, Sonora, Nayarit, Jalisco, and Oaxaca for the reference year 2022. The inventory follows the **GHG Protocol Corporate Standard** (WRI & WBCSD, 2004) and uses emission factors from the **IPCC 2006 Guidelines for National GHG Inventories**, with Global Warming Potential values from the **IPCC Sixth Assessment Report (AR6, 2021), GWP100 basis**.

The inventory covers Scope 1 (direct emissions from fuel combustion in fishing vessels), Scope 2 (indirect emissions from electricity at port ice plants and cold storage), and selected Scope 3 categories: Category 3 (upstream fuel) and Category 9 (downstream transportation and distribution of seafood products).



Key Findings

Key Finding: Scope 1 (fuel combustion) represents 75.2% of total fleet emissions (~605 kt CO₂e/year). Gasoline combustion in outboard motors is the primary hotspot. Sinaloa and Sonora together account for ~80% of Scope 1 due to larger fleets, higher-HP engines (75 HP), and more days at sea (165–180 days/year).

Key Finding: Scope 2 (electricity) is marginal at 2.2% (~17.5 kt CO₂e/year). Electrifying port ice plants, while beneficial, would not materially reduce fleet-wide emissions. The priority intervention is the fleet itself, not land-based infrastructure.

Key Finding: Air freight of premium shrimp (Category 9, Scope 3) generates ~43,400 tCO₂e from only 12,000 t of product (~4% of transported volume), about 80% of all Category 9 emissions. Per tonne-km, air freight is roughly 5–10 times more intensive than refrigerated trucking (1.13 vs. 0.115–0.21 kgCO₂e/t-km), and the 3,200 km average distance raises its footprint to ~3.6 tCO₂e per tonne of product. This makes air freight a priority target for Scope 3 reduction.

Key Finding: Scenario analysis indicates that transitioning to electric outboard motors could reduce total fleet emissions by ~73% (from 805 to ~219 kt CO₂e/year), combining an 80% cut in Scope 1 with the proportional drop in upstream fuel emissions (Cat. 3). Electricity for battery charging is not modeled. Efficient 4-stroke combustion engines offer a more near-term reduction of ~27%.

Important note on data and methodology: This inventory is a Tier 2 approximation based on publicly available data from CONAPESCA, FAO, SENER, and IPCC. Activity data for ice plants and product distribution are estimates derived from sector literature -- they do not represent direct measurements or verified records. Results should be interpreted as indicative of orders of magnitude and emission hotspots, not as a certified or audited inventory. A rigorous inventory would require primary data from fleet operators, energy bills, and logistics records. See Section 4 for full discussion of limitations.

1. Methodology

1.1 Accounting Standard and System Boundaries

The inventory follows the **GHG Protocol Corporate Accounting and Reporting Standard** (WRI & WBCSD, 2004, revised edition). The operational boundary applies the **operational control approach**, encompassing all emission sources over which the fleet exercises direct operational authority. The organizational boundary covers artisanal fishing vessels (*embarcaciones menores*) registered in the five Pacific coast states according to CONAPESCA 2022.

1.2 Greenhouse Gases and GWP Values

Three greenhouse gases are quantified: carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O). All gases are expressed as CO₂ equivalents (CO₂e) using GWP100 values from **IPCC AR6, Working Group I (2021), Chapter 7**: CH₄ = 27.9 (methane-only value from the chapter's Supplementary Material) and N₂O = 273 (Table 7.15). Using the AR6 fossil-methane value (29.8, Table 7.15) instead would raise total emissions by less than 0.1%.

1.3 Emission Factors

Emission Source	EF Value	Unit	Source Confidence
Scope 1 - CO ₂ (gasoline)	2.2620	kg CO ₂ /L	IPCC 2006, Vol.2, Ch.3, Table 3.2.1 High
Scope 1 - CH ₄ (gasoline)	0.3300	g CH ₄ /L	IPCC 2006, Vol.2, Ch.3, Table 3.3.1 High
Scope 1 - N ₂ O (gasoline)	0.0058	g N ₂ O/L	IPCC 2006, Vol.2, Ch.3, Table 3.3.1 High
Scope 2 - Grid Mexico (SEN)	0.4321	tCO ₂ e/MWh	SENER, Factor de Emisión del SEN, 2022 High
Scope 3 - Upstream gasoline	0.4800	kgCO ₂ e/L	IPCC 2006 + LCA literature, indicative Medium
Scope 3 - Truck <3.5t (refrig.)	0.2100	kgCO ₂ e/t·km	IPCC 2006, Vol.2 + EcoTransIT, indicative Medium
Scope 3 - Truck >3.5t (refrig.)	0.1150	kgCO ₂ e/t·km	IPCC 2006, Vol.2 + EcoTransIT, indicative Medium
Scope 3 - Air freight	1.1300	kgCO ₂ e/t·km	IPCC 2006, Vol.2 + EcoTransIT, indicative Medium

Emission factors marked as 'indicative' are approximations from secondary literature. Verification against primary sources (Ecoinvent, PEMEX data, operator records) is recommended for certified inventories.

1.4 Activity Data Sources

Fleet size and vessel characteristics derive from **CONAPESCA Anuario Estadístico de Acuacultura y Pesca 2022**. Fuel consumption rates per engine horsepower are based on **FAO Fisheries Technical Paper 484 (2008)**, which reports empirical consumption rates for outboard engines in small-scale fisheries. Days at sea per state are estimates derived from regional fishing calendars in CONAPESCA reports -- they are not directly measured values. Activity data for ice plants and cold storage are approximations based on FAO and SAGARPA sector reports and carry higher uncertainty than fleet combustion data.

1.5 Scope 3 Methodological Notes

Category 3 (Upstream fuel and energy): Fuel-based method. The upstream emission factor for gasoline (0.48 kgCO₂e/L) represents lifecycle emissions from crude extraction, refining, and transport to retail. This is an indicative value from Latin American LCA literature; verification against Ecoinvent 3.x or PEMEX-specific data is recommended for greater accuracy. Only upstream emissions of fuels are estimated; upstream emissions of purchased electricity (e.g., transmission and distribution losses) are not included.

Category 9 (Downstream transportation): Distance-based method (mass × distance × emission factor). Product volumes, routes, and modal split are geographic estimates based on CONAPESCA production data. These have not been validated against actual logistics records from operators or distributors.

2. Results

2.1 Scope 1 -- Direct Emissions from Fuel Combustion

Scope 1 emissions total approximately **605 kt CO₂e/year**, representing 75.2% of the fleet's total GHG footprint. All Scope 1 emissions originate from gasoline combustion in outboard motors. CO₂ accounts for more than 99% of Scope 1; CH₄ and N₂O together contribute less than 1%, consistent with gasoline combustion profiles (IPCC 2006, Vol.2, Ch.3). Sinaloa leads with ~289 kt due to its large fleet (8,420 vessels), high engine power (75 HP), and 180 days at sea per year. Oaxaca shows the lowest emission intensity (12 tCO₂e/vessel/year), reflecting 40 HP engines and only 120 days at sea.

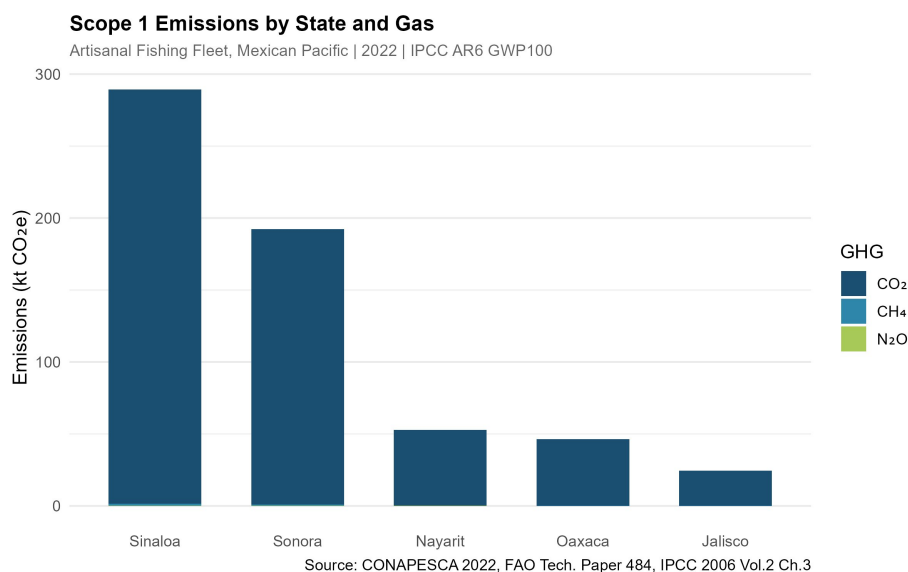


Figure 1. Scope 1 GHG emissions by state and gas species (kt CO₂e). CO₂ dominates across all states (>99%). Source: CONAPESCA 2022, FAO Technical Paper 484 (2008), IPCC 2006 Vol.2 Ch.3.

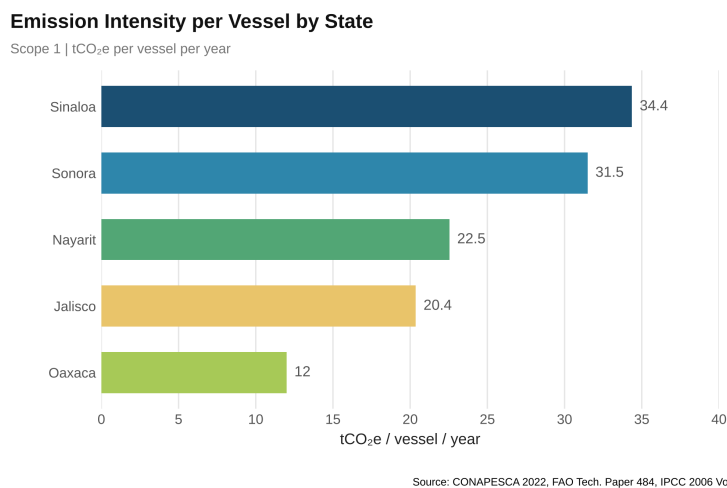
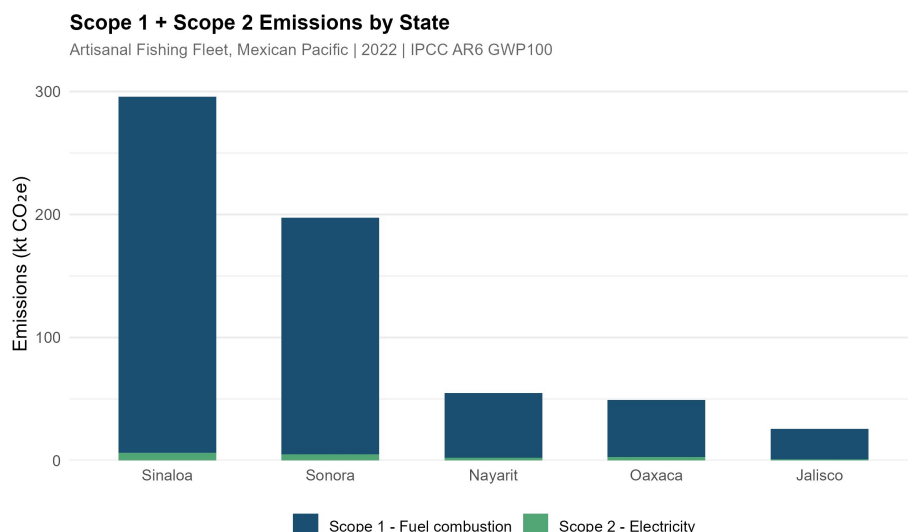


Figure 2. Scope 1 emission intensity per vessel (tCO₂e/vessel/year). North-south gradient reflects engine HP and fishing effort differences. Source: CONAPESCA 2022, FAO Technical Paper 484 (2008), IPCC 2006 Vol.2 Ch.3.

2.2 Scope 2 -- Indirect Emissions from Electricity

Scope 2 emissions total approximately **17.5 kt CO₂e/year** (2.2% of total), calculated using the location-based method with the national grid emission factor for Mexico (**SENER, Factor de Emisión del Sistema Eléctrico Nacional 2022: 0.4321 tCO₂e/MWh**). These emissions are associated with port ice plants and cold storage. Activity data are estimates with medium-to-high uncertainty and should be validated with direct metering data.

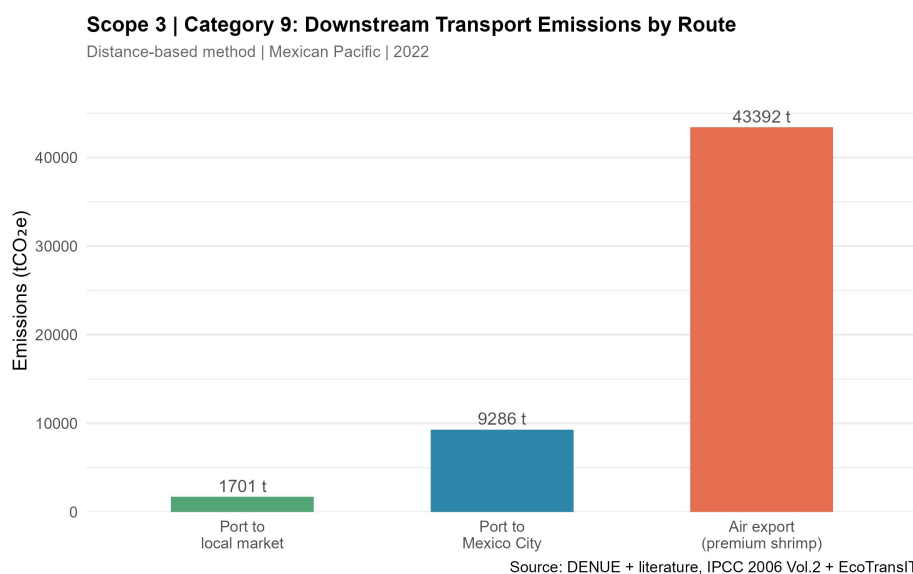


Source: CONAPESCA 2022, FAO Tech. Paper 484, SENER SEN 2022, IPCC 2006 Vol.2 Ch.3

Figure 3. Scope 1 vs. Scope 2 emissions by state (kt CO₂e). Scope 2 (green) is negligible relative to Scope 1, confirming port electricity is not a priority intervention point. Source: CONAPESCA 2022, SENER SEN Factor 2022, FAO Tech. Paper 484.

2.3 Scope 3 -- Value Chain Emissions

Total Scope 3 emissions are approximately **182 kt CO₂e/year** (22.6% of total). **Category 3 (upstream fuel):** ~128 kt, proportional to Scope 1 fuel consumption, calculated with an indicative upstream factor of 0.48 kgCO₂e/L for gasoline. **Category 9 (downstream transport):** ~54 kt in total, of which ~80% (43 kt) comes from air freight of premium shrimp despite representing only 12,000 t of product (~4% of transported volume). Per tonne-km, air freight is roughly 5–10 times more intensive than refrigerated trucking (1.13 vs. 0.115–0.21 kgCO₂e/t·km); over the 3,200 km average route it emits ~3.6 tCO₂e per tonne of product, compared with ~0.1 tCO₂e/t on the Mexico City route. This disproportionate impact highlights air freight as a high-priority Scope 3 reduction opportunity.



Source: DENUe + literature, IPCC 2006 Vol.2 + EcoTransIT

Figure 4. Scope 3 Category 9 downstream transport emissions by route (tCO₂e). Air export of premium shrimp generates ~43,400 tCO₂e from 12,000 t of product -- about 80% of Category 9 emissions, and roughly 5–10 times more per tonne-km than refrigerated truck distribution. Source: Geographic estimates (DENUe + literature), IPCC 2006 Vol.2, EcoTransIT. Note: product volumes and distances are approximations.

3. Consolidated Inventory and Scenario Analysis

3.1 Consolidated GHG Inventory -- All Scopes

Scope	Emission Source	tCO ₂ e/yr (approx.)	% Total	Data Confidence
Scope 1	Fuel combustion (outboard motors)	~605,000	75.2%	High
Scope 2	Electricity (ice plants & cold storage)	~17,500	2.2%	Medium
Scope 3 Cat.3	Upstream fuel production	~128,000	15.9%	Medium
Scope 3 Cat.9	Downstream transportation	~54,400	6.8%	Low-Medium
TOTAL	All scopes	~805,000	100%	--

Table 1. Consolidated GHG inventory by scope. Values are Tier 2 estimates. Confidence: High = official statistics + published EFs; Medium = sector literature estimates; Low-Medium = geographic approximations.

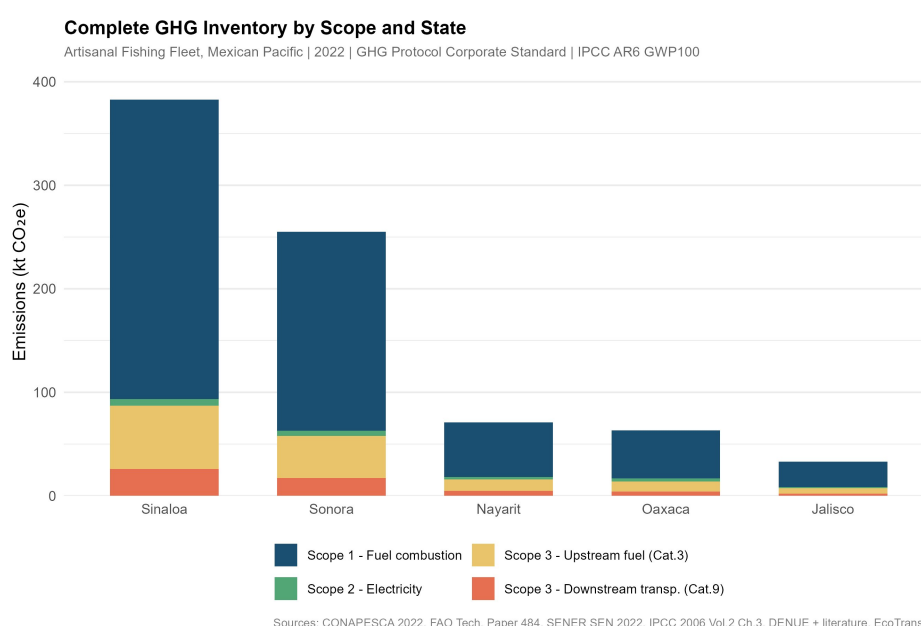
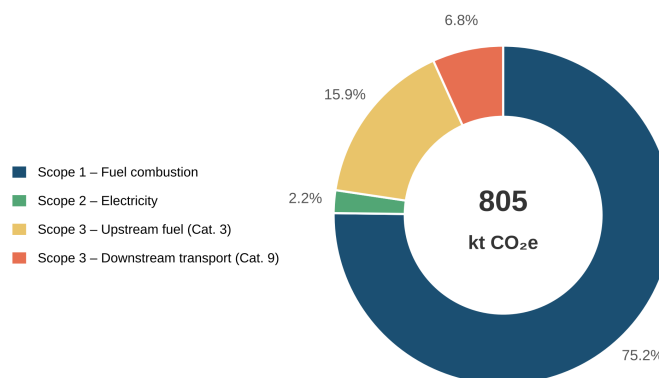


Figure 5. Complete GHG inventory by scope and state (kt CO₂e). Sinaloa and Sonora account for ~79% of total fleet emissions. Source: CONAPESCA 2022, FAO Tech. Paper 484, SENER SEN 2022, IPCC 2006 Vol.2, DENU + literature, EcoTransIT.

Fleet-wide GHG Emissions by Scope

Artisanal Fishing Fleet, Mexican Pacific | 2022

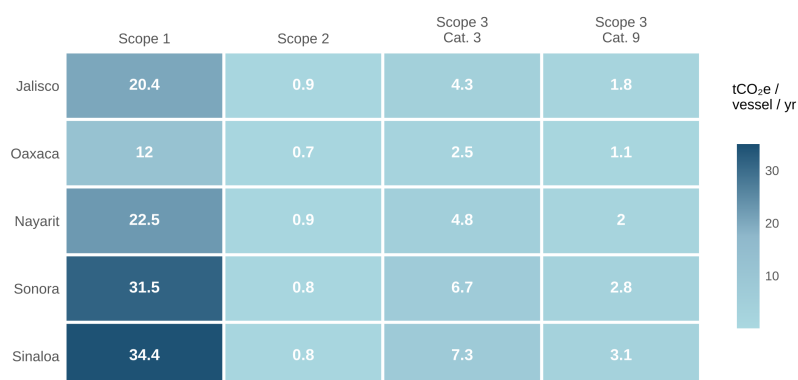


GHG Protocol Corporate Standard | IPCC AR6 GWP100

Figure 6. Fleet-wide GHG emissions by scope (% of 805 kt CO₂e total). Scope 1 fuel combustion = 75.2%. GHG Protocol Corporate Standard | IPCC AR6 GWP100.

Emission Intensity per Vessel by Scope and State

tCO₂e per vessel per year | GHG Protocol Corporate Standard | IPCC AR6 GWP100



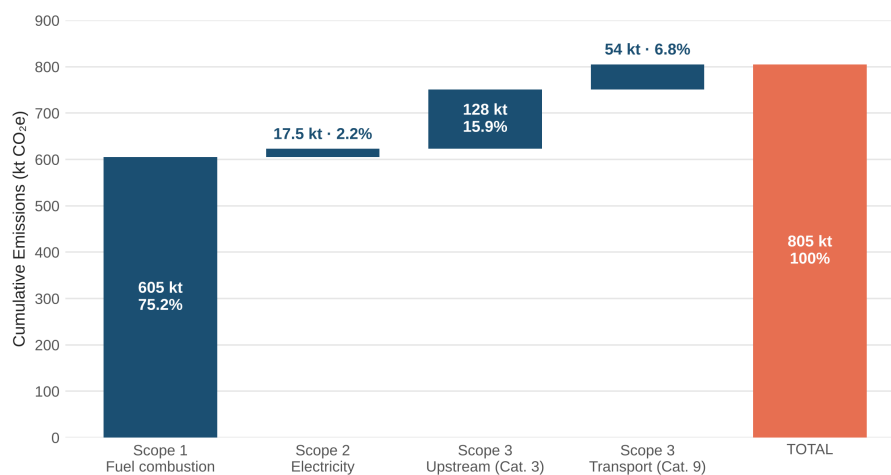
Sources: CONAPESCA 2022, FAO Tech. Paper 484, SENER SEN 2022, IPCC 2006, EcoTransIT

Figure 7. Emission intensity heatmap (tCO₂e/vessel/year) by scope and state. Scope 1 shows the strongest geographic gradient. Scope 2 is uniformly low (<1 tCO₂e/vessel/year) across all states. Source: CONAPESCA 2022, FAO Tech. Paper 484, SENER SEN 2022, IPCC 2006.

3.2 GHG Emissions Waterfall

GHG Emissions Waterfall: Contribution by Scope

Artisanal Fishing Fleet, Mexican Pacific | 2022 | Total: 805 kt CO₂e



GHG Protocol Corporate Standard | IPCC AR6 GWP100

Figure 8. Cumulative GHG emissions waterfall by scope (kt CO₂e). Each bar shows the incremental contribution of each scope to the 805 kt CO₂e total. GHG Protocol Corporate Standard | IPCC AR6 GWP100.

3.3 Decarbonization Scenario Analysis

Four illustrative scenarios assess the potential magnitude of different intervention strategies. These are indicative only and do not account for implementation costs, technical feasibility in artisanal fishing contexts, or transition timelines. They identify priority intervention areas, not targets or commitments.

Scenario	Key Assumption	Total tCO ₂ e	vs. Baseline
Baseline 2022	Current fleet -- gasoline outboard motors	~805 kt	0%
Efficient motors (-30%)	30% fuel reduction via modern 4-stroke engines; Scope 1 and Cat. 3 fall proportionally	~585 kt	-27%
Electric outboards (-80%)	80% Scope 1 reduction via electric propulsion; Cat. 3 falls proportionally; Cat. 9 unchanged	~219 kt	-73%
Grid decarbonization	60% Scope 2 reduction via renewables (Mexico NDC)	~795 kt	-1.3%
Combined best case	Electric outboards + grid decarbonization	~208 kt	-74%

Table 2. Decarbonization scenarios. Assumptions: efficient motors = 30% fuel reduction (4-stroke vs. 2-stroke technology literature); electric outboards = 80% Scope 1 reduction; grid decarbonization = 60% Scope 2 reduction aligned with Mexico NDC renewable energy targets. Upstream fuel emissions (Cat. 3) are assumed to scale with fuel use; downstream transport (Cat. 9) is unchanged. Electricity for charging batteries is not modeled; including it would add Scope 2 emissions and reduce the net benefit of electrification. All scenarios are illustrative estimates.

GHG Reduction Scenario Analysis

Potential emission reductions under different decarbonization pathways | 2022 baseline

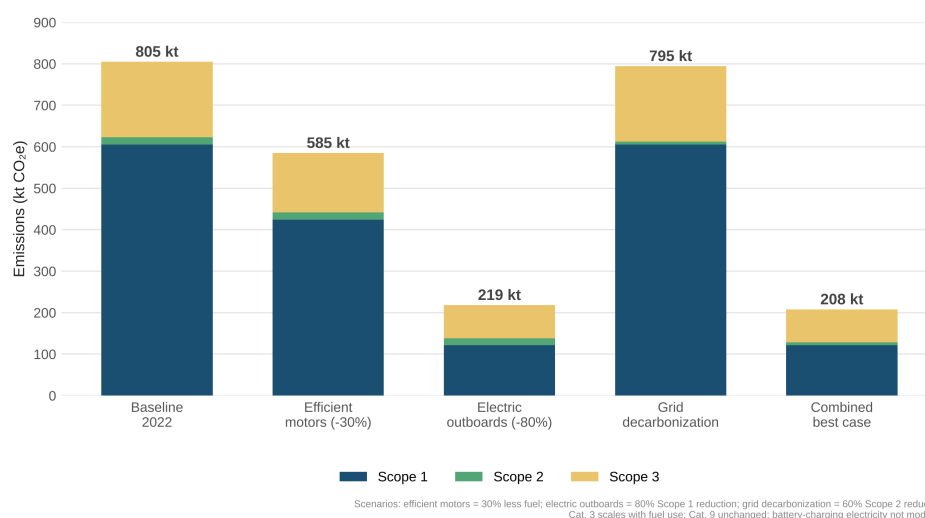


Figure 9. GHG reduction scenario analysis (kt CO₂e). Electric outboard motors represent the highest-impact single intervention (-73%). Grid decarbonization alone has minimal fleet-wide impact given Scope 2's 2.2% share of total emissions.

4. Limitations and Uncertainty

Transparent disclosure of data quality and methodological limitations is a core principle of the GHG Protocol Corporate Standard. The following limitations apply to this inventory and must be considered when interpreting results.

- **Tier 2 approximation -- not a verified inventory.** This report uses secondary data and published emission factors. It has not been subject to third-party verification and must not be used for regulatory compliance, carbon credit issuance, or certified reporting without independent audit (ISO 14064-3).
- **Fleet activity data uncertainty.** Days at sea per state and engine operating hours are estimated from CONAPESCA fishing calendars -- not directly measured. Actual fuel consumption may vary significantly with sea conditions, vessel maintenance, and seasonal patterns. A $\pm 20\text{--}30\%$ uncertainty range in Scope 1 results is plausible.
- **Ice plant and cold storage data.** The number of facilities and their electricity consumption are approximations based on FAO and SAGARPA sector reports. No direct census of port infrastructure was conducted. Scope 2 results should be treated as order-of-magnitude estimates only.
- **Scope 3 upstream emission factor.** The gasoline upstream factor ($0.48 \text{ kgCO}_2\text{e/L}$) is derived from generic Latin American LCA literature. Mexico-specific data from PEMEX refining operations or Ecoinvent 3.x would substantially improve Category 3 accuracy.
- **Scope 3 Category 9 distribution data.** Product volumes, routes, and modal split are geographic estimates not validated against supply chain records. Actual logistics may differ substantially, particularly for export volumes and air freight.
- **Scenario analysis.** Scenarios are simple proportional reductions of the baseline. Electricity needed to charge electric outboards is not included, so the electrification results are an upper bound on the achievable reduction.
- **Single reference year.** Fleet composition, fishing effort, and grid emission factors change over time. This inventory represents a 2022 snapshot and does not capture interannual variability.

Recommended next steps for a rigorous inventory:

- Obtain primary fuel consumption data from fleet operators and CONAPESCA fishing logbooks (*bitácoras de pesca*).
- Survey port ice plants and cold storage for actual electricity consumption (CFE billing records).
- Validate Scope 3 upstream factors against Ecoinvent 3.x for Mexican petroleum refining.
- Collect actual logistics data from seafood distributors and export operators for Scope 3 Category 9.
- Quantify battery-charging electricity demand to complete the electric-outboard scenario.
- Commission third-party verification following ISO 14064-3 for a certifiable inventory.

5. References

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- [7] **IPCC (2021)**. Climate Change 2021: The Physical Science Basis. Working Group I, Sixth Assessment Report. Chapter 7 (Table 7.15) and Supplementary Material: GWP100 values. Cambridge University Press.
- [8] **SENER (2022)**. Factor de Emisión del Sistema Eléctrico Nacional 2022. Secretaría de Energía, Mexico.
- [9] **EcoTransIT World (2023)**. Methodology and Data. EcoTransIT World Initiative. Available at: ecotransit.org. Note: transport emission factors used in this report are indicative values based on EcoTransIT methodology; direct tool verification is recommended for certified inventories.

This report was produced as part of a professional portfolio project demonstrating applied GHG accounting methodology for the fisheries sector. All analyses were conducted in R (tidyverse, ggplot2) using publicly available data.